INCH-POUND

A-A-59257A
July 14, 2004
SUPERSEDING
A-A-59257
May 19, 1998

COMERCIAL ITEM DESCRIPTION
OSCILLOSCOPE
(Digital Storage, 500 MHz)

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE.

This CID outlines the minimum requirements for a 500-Megahertz (MHz), four-channel oscilloscope with a two Giga-sample per second (GS/sec) sampling rate and digital storage, with a differential probe for making differential measurements. This commercially available digital storage oscilloscope will be used on a bench and is capable of being modified with a rack-mount kit to be included in a rack-mount assembly. This CID is meant as a minimum requirement for the oscilloscope, and only those manufacturers that meet or surpass the following requirements are to supply the oscilloscope per this CID.

2. SALIENT CHARACTERISTICS.

2.1 Equipment covered by this CID shall be commercially available equipment and may modified to the extent necessary to meet the following description. The equipment shall be Class 4, in accordance with MIL-PRF-28800. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.2 Safety and Environmental. The oscilloscope shall meet all safety and environmental requirements as specified in MIL-PRF-28800 when used as bench-top or rack-mount equipment.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database t www.dodssp.daps.mil.

AMSC N/A FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

2.3 Electrical power sources. The equipment power consumption of all components combined shall not exceed 400 Volt amps and the oscilloscope shall operate from nominal commercial military, and shipboard power sources over the following ranges: at 110/120 Volts rms (± 10 percent), single phase at frequencies of 50, 60, and 400 Hertz (Hz) (± 5 percent) and 200/240 Volts rms (± 10 percent) single phase at line frequencies of 50 and 60 Hz (± 10 percent). The oscilloscope shall operate in accordance with the steady state conditions, transient state conditions, and interruption of power source requirements outlined in MIL-PRF-28800.

2.4 Accessories. Accessories to be supplied by the manufacturer with each deliverable oscilloscope shall include standard accessories such as the appropriate power cord(s), probe(s), accessory pouch, front panel cover, a written tutorial/user reference manual, a written programmer reference manual, and a traceable calibration certificate.

2.4.1 Cover. The oscilloscope enclosure shall have a cover protecting the front panel controls and display in accordance with MIL-PRF-28800.

2.4.2 Probes. The active or passive probes provided for all channels as standard accessories or to meet the requirements contained herein shall meet the accuracy requirements over the full 500 MHz bandwidth and vertical accuracy of the oscilloscope. The probe attenuation shall be automatically recognized and the scale factor of the oscilloscope adjusted appropriately.

2.4.3 Differential Probe. Each DSO shall come with a differential probe that has a 500 MHz bandwidth and greater than 40 dB CMRR (Common Mode Rejection Ratio) at 10 MHz. This differential probe shall have the flexibility of 10X gain and 10:1 attenuation for a maximum of 100:1 attenuation.

2.5 Interface. The oscilloscope shall be capable of interfacing with industry standard data interchange formats for files, formats, storage, communications, plotting, and printing.

2.5.1 ANSI/IEEE-STD-488 Interface Requirements. The oscilloscope's digital interface shall control each device function including all ranges, modes, and levels except for primary power control and other non-measurement functions. Requirements shall be as stated in MIL-PRF-28800, Section 3.

2.5.2 Output Ports. The oscilloscope shall have RS-232 interface that complies at a minimum with talk only commands and Centronics interface complying with printer and/or plotter outputs.

2.5.3 Printer Interface. The oscilloscope shall have the minimum capability of interfacing with Laser and Epson style printers.

2.5.4 File Formats. The oscilloscope shall have, at the least, capability of providing, storing, and transferring data in TIFF and BMP image formats.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

2.5.5 Data Format The oscilloscope shall have the ability of saving data in ASCII output compatible with MathCAD™ and spreadsheet formats.

2.6 Dimensions and Weight. The oscilloscope dimensions shall permit the oscilloscope's inclusion in a standard 19-inch wide rack enclosure with a standard height of less than 6 rack units. The oscilloscope's weight, including the front cover and all accessories (excluding written manuals) shall not exceed 48 pounds.

2.7 Self-Test. The oscilloscope shall provide an automatic diagnostic self-test, in accordance with MIL-PRF-28800, Section 3, automatically upon start-up and/or when invoked by the operator either manually or remotely.

2.8 Extended Diagnostics. The oscilloscope shall provide extended diagnostic routines to detect internal electrical failures to the next lower level from self-test and shall be invoked by front panel controls.

2.9 Signal Path Compensation. The oscilloscope shall have the capability to ensure specified DC and timing accuracy. Changes in environmental conditions within the operating range specified for the equipment will not require additional calibration of the oscilloscope, except for internal calibration procedures and signal path compensation, to meet performance requirements. The internal calibration may be initiated automatically or manually.

2.10 Calibration. The oscilloscope shall meet the requirements for calibration in MIL-PRF-28800F.

2.11 Autoset. The oscilloscope shall provide an Autoset function that when activated, will automatically configure the vertical, horizontal, triggering, acquisition, and display controls to provide a triggered display positioned on a screen.

2.12 Temperature Requirements. The oscilloscope shall meet performance and accuracy requirements in Class 4 operating and non-operating environments.

2.13 Humidity. The oscilloscope shall meet performance and accuracy requirements for Class 4 Humidity of MIL-PRF-28800 under non-condensing conditions.

2.14 Shock, Functional. The oscilloscope shall meet its performance and accuracy requirements in an operating environment after being subjected to a 20-g half-sine shock test.

2.15 Altitude. The maximum operating altitude shall be 2000 meters, and the oscilloscope will be functional after being transported at a maximum altitude of 12,000 meters.

3. PERFORMANCE CHARACTERISTICS.

3.1. Warm-up Performance. The equipment shall perform to specifications after a warm-up period not to exceed thirty minutes after initiating power and all calibration and self-check routines have been accomplished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.2. Acquisition System. This digitizing oscilloscope shall acquire data on all channels simultaneously and will capture waveforms in a single trigger event up to the full rated 500 MHz bandwidth on at least four channels.

3.2.1. Sample Mode. The oscilloscope shall provide a sample mode in real time and/or repetitive sampling which may be sequential, random, interleaved or any other technique to accomplish a reconstruction of the waveform.

3.2.2. Dead Band or Dead Time Between Segments. The oscilloscope's dead band or dead time between waveform captures shall be I millisecond or less for one channel acquisition, triggered sweep mode, math off, interpolation on and vectors on.

3.2.3. Bandwidth. The oscilloscope bandwidth specifications of 500 MHz shall apply for all channels when coupled at 50 ohm or 1 Megohm unless otherwise specified herein. The oscilloscope bandwidth specification applies with probes installed.

3.2.3.1. DC Coupled Bandwidth. The DC coupled frequency response for the 50 ohm input shall not be more than +1.5 dB up or more than -3 dB down from DC to 500 MHz with or without the probe attached for volts/division settings of 1 millivolt and above. The DC coupled frequency response for the 1 Megohm input shall not be more than +1.5 dB up or more than -3 dB down from DC to 500 MHz with or without the probe attached for volts/division settings of 1 millivolt and above.

3.2.3.2. AC Coupled Bandwidth. The AC coupled frequency response for the 1 Megohm input shall not be more than +1.5 dB up or more than -3 dB down from 10 Hz to 500 MHz with or without the probe attached.

3.2.3.3. Bandwidth Limiting. The oscilloscope shall provide the capability to limit the upper bandwidth of all channels to 20 MHz and full bandwidth.

3.2.3.4. Step Recovery Settings. The oscilloscope shall provide short-term settling time that recovers with less than 1 percent error within 60 microseconds or less when subjected to a 2 volt step input with a rise time greater than 175 picoseconds when tested in accordance with paragraph 4.8 of IEEE Std 1057-1994.

3.3. Sample Rate. The oscilloscope shall have a minimum sample rate for real time sampling of 4 GS/s for one or two channel operation and a minimum of 2 GS/s with three or more channels in operation.

3.4. Record Length. The oscilloscope shall be capable of acquiring a minimum of 2M samples simultaneously for each channel. Additional record lengths will be selectable as provided.

3.5. Acquisition Modes. The oscilloscope shall provide the following acquisition modes:

3.5.1. Sample. The oscilloscope shall sample data in accordance with an internal algorithm allowing data to be acquired dependent upon the time/division setting and record length selected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.5.2. Average. The oscilloscope shall sample data over multiple acquisitions and provide an average composite waveform of the data. The average range shall be selectable from 2 to 4096 waveforms.

3.5.3. Pulse Capture. The equipment shall capture pulses of at least 3 ηs in duration for the real time sample rate.

3.5.4. Single. The oscilloscope shall capture a single waveform in accordance with an internal algorithm allowing data to be acquired determined by the time/division setting and record length selected either manually or by a triggered input.

3.5.5. Increased Resolution. The oscilloscope shall have a mode, which will increase vertical resolution from 8 bits to a minimum of 10 bits of vertical resolution.

3.6. Horizontal or Time Base System. The oscilloscope shall provide the following horizontal timing functions:

3.6.1. Seconds/Division Range. The oscilloscope shall provide a seconds/division range of at least 1 ηs/division to 10 s/division in a 1-2-5 sequence.

3.6.2. Time Base Accuracy. The oscilloscope shall have a time base accuracy of at least ± 25 parts per million (ppm) over the temperature range of +15°C to 35°C.

3.6.3. Interpolator Resolution. The oscilloscope shall have a resolution of at least 10 picoseconds.

3.6.4. Horizontal Jitter. The oscilloscope shall have less than 100 picoseconds rms of horizontal jitter.

3.6.5. Delay Between Channels. The oscilloscope delay between channels shall not exceed 250 ρs for any channels with equal volts/division and coupling settings

3.7. Vertical System. The oscilloscope shall function as follows:

3.7.1. Input Coupling. The oscilloscope shall provide AC coupling and DC coupling for the 1 Megohm input and DC coupling for the 50 ohm input. A ground reference or on screen designation of a ground reference shall be provided.

3.7.2. Sensitivity Range. The oscilloscope shall provide sensitivity ranges from a minimum of 1 millivolt/division to at least 5 volts/division for the 1 Megohm input and from a minimum of 1 millivolt/division to at least 1 volt/division for the 50 ohm input. The sensitivity ranges shall be in a 1-2-5 sequence. Indicated sensitivities shall be automatically switched to display the correct scale factor when properly coded attenuator probes are attached.

3.7.3. Offset Range. The oscilloscope's offset range for all channels shall be a minimum of ±4 divisions for all Volts/division settings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.7.4. Position Range. The oscilloscope shall provide the capability to position each displayed trace from the center of the display to the top and bottom of the display.

3.7.5. Display Dynamic Range. The oscilloscope shall provide the capability to avoid clipping on varying signal artifacts that exceed the top and bottom display area. This may be accomplished by, but not limited to, auto setup features, display indications, warnings, increased dynamic range. These capabilities may be provided either individually or in combination.

3.7.6. Channel Isolation. The oscilloscope isolation between the input channels, with identical Volts/division settings, shall be at least 40 dB at 100 MHz and at least 30 dB at 500 MHz.

3.7.7. DC Gain Accuracy. The oscilloscope DC gain accuracy shall be at least ± 2.0% of full scale at 0 volts offset.

3.7.8. Input Impedance. The oscilloscope input impedance shall be selectable between 1 Megohm (±1.0%) and 50 Ohms (±1.5%). The VSWR for 50-ohm input shall be =1.5:1 from DC to 500 MHz.

3.7.9. Maximum Input Voltage. The oscilloscope shall withstand the following requirements without damage:

3.7.9.1. One Megohm Input Impedance. The oscilloscope inputs shall withstand, without damage, up to 250 volts (DC plus peak AC), de-rated at 20 dB/decade above 10 kHz when 1 Megohm input impedance selected.

3.7.9.2.50 Ohm Input Impedance. . The oscilloscope inputs shall withstand, without damage, up to 5 volts RMS with peaks less than or equal to ± 30 volts and not exceeding 1 microsecond in duration with 50 ohms input impedance selected.

3.7.9.3.Vertical Resolution. The oscilloscope shall have a minimum of 8 bits of resolution in real time sampling mode and a minimum of 10-bit resolution in repetitive sampling mode.

3.7.9.4. Effective Bits of Resolution. The oscilloscope shall have a minimum of 4.5 effective bits of resolution at one-fourth the maximum bandwidth when measured at 80% of full scale. Testing shall be accomplished from the lowest AC bandwidth in 10 MHz intervals up to one-fourth of maximum bandwidth. The highest sample rate and record size shall be selected in consonance with the internal algorithm of the oscilloscope time base and sample rate settings. The test frequencies shall be calculated in accordance with IEEE Standard 1057, Section 4.

3.7.10. Vertical Zoom. . The oscilloscope shall have the ability to vertically zoom up to 5X expansion or up to 50X expansion with averaging and accomplish this at a sensitivity of 50 µV/division.

3.8. Triggering System. The oscilloscope shall provide both main and delayed triggering, edge triggering, logic triggering, and pulse triggering capabilities. Edge triggering shall be fully configurable for source, slope, coupling, mode, and hold off.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.8.1. Triggering System Bandwidth. The oscilloscope shall provide triggering over the full 500 MHz bandwidth.

3.8.2. Time Base Trigger Modes. The oscilloscope shall provide time base trigger selectable in main and delay modes.

3.9. Main Trigger . The oscilloscope shall provide the following main trigger modes.

3.9.1. Normal Mode. The oscilloscope main time base trigger shall be selectable from a vertical amplifier signal, auxiliary signal, or the power line. A waveform shall be acquired only in the presence of a trigger signal. When no trigger signal occurs, there shall be no acquisition of a waveform. Loss of the trigger shall stop any multiple acquisition sequence in progress and freeze the display, regardless of the selected time/division setting.

3.9.2. Automatic Mode. The oscilloscope shall initiate acquisition when a trigger occurs or after a certain time period (that varies based on the time/division setting) following the end of the last acquisition, whichever occurs first. A waveform shall be displayed even though the oscilloscope is not triggered. Automatic mode uses a timer that starts after a trigger event occurs. If another trigger event is not detected before the timer runs out, the oscilloscope generates another trigger.

3.9.3. Single Mode. The oscilloscope shall initiate acquisition and shall run one time or in accordance with the acquisition mode selected following rearming for each selected vertical channel and sweep combination, when triggered as in the normal mode. A front-panel control shall rearm the sweep and illuminate a reset indicator.

3.10. Delay Trigger. The oscilloscope shall have a delayed trigger mode providing delay by time or delay by events.

3.10.1. Delay Jitter. The oscilloscope shall have a delay jitter ≤ 40 ppm of the time base delay setting.

3.10.2. Time Delay. The oscilloscope shall provide a trigger mode that delays the initiation of a display and timing sequence from 20 nanoseconds to 20 seconds.

3.10.3. Time Delay Accuracy. The oscilloscope delay time accuracy shall be identical to the main time base accuracy over the temperature range of +15°C to +35°C.

3.10.4. Delay By Events. The oscilloscope shall provide a trigger mode that delays the initiation of the display and timing sequence from 2 to 9,999,999 events. An event is characterized by a positive or negative transition through the trigger level as defined by the type of coupling.

3.11. Edge Triggering. The oscilloscope shall provide edge triggering in accordance with the following:

3.11.1. Trigger Level And Slope. The oscilloscope shall provide a continuously variable trigger level control. Slope controls shall also be provided to permit triggering on any level on the positive and negative slopes of a trigger signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.11.2. Trigger Level or Trigger Threshold Ranges. The upper and lower voltage limits that the trigger level or threshold can be set to are ± 4 divisions from screen center for any channel trigger source, ± 4 volts for auxiliary trigger source, and 250 volts maximum for line trigger source with slope selectable at a minimum.

3.11.3. Input Channel. The oscilloscope trigger signal is derived from any one of the primary inputs or trigger channel. If the trigger source is one of the primary channels selected for display, you can view the trigger source. You can also trigger from an input source not being displayed.

3.11.4. AC Line Voltage. The oscilloscope trigger signal is derived from the AC line voltage. AC line is available as a source only for the Main Time Base trigger.

3.11.5. External Trigger. The oscilloscope shall provide an external trigger input.

3.11.5.1 External Trigger Level Range The oscilloscope shall have a minimum trigger level range of ± 500 millivolts.

3.11.5.2 External Trigger Level Edge Sensitivity. The oscilloscope shall trigger on a minimum of 400 mV up to 50 MHz and a minimum of 750 mV up to 100 MHz or 0.05 X signal amplitude (to 100MHz) , 0.1 X signal amplitude (to 500MHz).

3.11.5.3 External Trigger Level Maximum Input. The oscilloscope shall accept external inputs up to ± 20V (DC + peak AC).

3.11.5.4 External Trigger Bandwidth. The oscilloscope external bandwidth shall be at least DC to 500 MHz.

3.11.6. Edge Type Trigger Coupling. The oscilloscope shall provide the following trigger coupling modes:

3.11.6.1 Direct Current Trigger (DC). The oscilloscope shall trigger on both AC and DC signal components when signal levels exceed 1.0 division to 50 MHz and 2.0 divisions to 500 MHz.

3.11.6.2 Alternating Current Trigger (AC). The oscilloscope shall trigger on the AC signal components above approximately 60 Hz when signal levels exceed 1.0 division to 50 MHz and 2.0 divisions to 500 MHz.

3.11.6.3 High Frequency Rejection. The oscilloscope shall trigger on signals at 1.0 division from DC to approximately 30 kHz.

3.11.6.1 Low Frequency Rejection. The oscilloscope shall trigger when signals exceed 1.0 division above 80 kHz to 50 MHz and 2.0 divisions to 500 MHz.

3.11.7 Trigger Position. The position of the trigger in the waveform record shall be selectable in 1% increments to allow 0 to 100% of the data to be acquired.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.11.8 Trigger Jitter. The oscilloscope trigger jitter shall not exceed 100 picoseconds rms at full bandwidth and the fastest sweep setting.

3.11.9 Pulse Triggering. The oscilloscope shall provide triggering on pulse edges with the minimum characteristics described herein.

3.11.9.1 Glitch Triggering. The oscilloscope shall trigger or reject signals on the rising or falling edges of pulses with widths at least over the range of 5 ηs to 1 second.

3.11.9.2 Glitch Triggering Time Accuracy. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 5 nanoseconds to 1 microsecond, of ± (20 percent of setting + 1.5 nanoseconds) and for a time range greater than 1 microsecond and less than or equal to 1 second; accuracy shall be ± 1.0 percent.

3.11.9.3 Pulse Width Triggering. The oscilloscope shall trigger on a positive or negative pulse within or not within the selectable time limits of at least 2.5 ηs to 1 second.

3.11.9.4 Time Accuracy For Pulse Width Triggering. The oscilloscope shall have a minimum time accuracy for pulse width triggering for a time range of 2.5 nanoseconds to 1 microsecond, of ± (20 percent of setting + 1.5 nanoseconds), and for a time range greater than 1 microsecond; accuracy shall be ± 1.0 percent measured at 50 percent.

3.11.10 Time-out Triggering. The oscilloscope shall trigger when a pulse fails to complete a transition from a high to low state and return to the original state within the time range of at least 25 ηs to 10 seconds.

3.11.11 Video Triggering. The oscilloscope shall provide video triggering in PAL for 625 lines/50 Hz, NTSC for 525 lines/60 Hz and non-standard video formats as described herein in monochrome or color.

3.11.11.1 Video Trigger Selection. . The oscilloscope shall trigger on the TV input signal referenced to line or field number or both.

3.11.11.2 Video Trigger Sensitivity. The oscilloscope shall trigger on 0.8 divisions of composite SYNC signal.

3.11.11.3 Non-Standard Video Triggering. The oscilloscope shall provide a triggering capability for non-standard video by providing the ability to trigger on any line up to 1500 and odd or even fields.

3.12 Magnification. The oscilloscope shall provide a horizontal magnification adjustable over a minimum range of 1 to 1000. The oscilloscope shall position waveforms both vertically and horizontally and read out on screen of the vertical and horizontal scaling factors.

3.13 Measurement System Features. The oscilloscope shall provide the following measurement capabilities:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.13.1 Automated Measurements. The oscilloscope shall provide automated measurements defined by IEEE-STD-181-1977. The oscilloscope measure function will automatically extract parameters from any signal input to the scope. Four or more of the parameters can be displayed on screen and made available for transfer over the IEEE-488 interface bus. The parameters measured shall include, but are not limited to: period, frequency, width, rise time, fall time, duty cycle, delay, positive overshoot, negative overshoot, maximum, minimum, peak-to-peak, amplitude and root mean square (rms).

3.13.2 Automated Measurement Accuracy Voltage. The oscilloscope shall have a minimum accuracy of ± 4 percent of full scale for voltage related measurements for single point measurements and ± 2 percent of full scale for dual point voltage measurements with the signal at 50% or higher of full scale.

3.13.3 Automated Measurement Accuracy Time. The oscilloscope shall have minimum accuracy of ± 1 percent measurements for dual point time-related measurements.

3.13.4 Cursor Measurements. The oscilloscope shall have the capability to make single and dual cursor measurements along both the vertical and horizontal axis with and without tracking. Cursors shall be provided for making vertical and horizontal measurements on the displayed waveforms. The cursors shall have the ability to measure absolute and delta voltage between the positions of the horizontal bar cursors and absolute and delta time in seconds or Hz for 1/time between the positions of the vertical bar cursors. The time and voltage related values represented by cursor positions shall be displayed on screen and available for transfer over the IEEE-488 interface bus.

3.13.5 Cursor Measurement Accuracy Voltage. The oscilloscope shall have a minimum accuracy of ± 4 percent of full scale for single cursor voltage related measurements and ± 2 percent of full scale for dual cursor voltage related measurements. Accuracy requirements pertain to stable waveforms.

3.13.6 Cursor Measurement Accuracy Time. The oscilloscope shall have a minimum accuracy of ± 1 percent of full scale for time related measurements

3.13.7 Cursor Readout. The oscilloscope shall provide a cursor readout selectable by the user to display cursor measurement in either voltage, time, or frequency or any combination of these measurements. Dual cursor measurements shall provide a readout of the relative positions of the cursors and the difference.

3.14 Math Functions. The oscilloscope shall provide math waveform operations of addition, subtraction, inversion, and multiplication of any two waveforms in any combination from the current displayed waveforms or stored waveforms.

3.15 Display, Controls, & Adjustments. The oscilloscope shall provide the following display, controls and adjustments:

3.15.1 Display. A color display or visual device capable of displaying both waveforms in normal ambient room light in sufficient detail for direct measurements on screen without

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

automated measurements or cursors. The minimum viewing area of the display shall be at least 10 centimeters (cm) horizontally and 8 cm vertically.

3.15.2 Display Style. The oscilloscope shall provide waveform displays in sin (x)/x, linear, dots, variable persistence, and infinite persistence styles.

3.15.3 Trace Convergence. The oscilloscope shall have the capability to discriminate and display where multiple waveforms converge.

3.15.4 Color Display. The oscilloscope shall have a color display with the capabilities described herein.

3.15.4.1 Color Conventions. The oscilloscope color display shall exhibit at least seven colors of the visible light spectrum, which are red, orange, yellow, green, blue, indigo, and violet.

3.15.4.2 Color Grading. The oscilloscope shall have color grading and exhibit such when the variable persistence mode is selected. The waveform points shall time decay through a minimum of 12 levels of intensity.

3.15.4.3 Color Resolution. The oscilloscope shall have the resolution of a standard VGA monitor, or better such that the resolution is 640 horizontal points by 480 vertical points.

3.16 Display Intensity. The oscilloscope shall provide the capability to control overall intensity, text/graticule intensity, and waveform intensity.

3.17 Display Persistence. The oscilloscope shall provide both infinite and variable persistence. The variable persistence range shall be over the range of 500 ms to 10 seconds.

3.18 Display Intensity or Color Scaling. The oscilloscope shall provide the ability to provide at least 8 discernable intensity levels.

3.19 Display Graticule Size and Markings. The oscilloscope performance shall be over a minimum 8 X 10-division graticule in the waveform display area. The cardinal axes shall be ruled in 0.2 minor divisions.

3.20 Display Graticule Types. The oscilloscope shall provide the capability of selecting at least full screen, XY display, or grid displays.

3.20.1 Waveform Display Area. The oscilloscope total waveform display area shall have at least 640 horizontal points by at least 480 vertical points. Actual viewing area shall be at least 500 horizontal points by at least 400 vertical points.

3.20.2 Screen Data Displayed. Horizontal and vertical scaling factors for each channel, time base, trigger sources, trigger level, trigger type, sample rate, sampling mode, at least four measurements, and a representation of memory length with trigger point indicated.

3.21 Format. The oscilloscope shall have both YT and XY axis formats.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.22 "Fit to" and "Fill" The Screen. The oscilloscope shall have the ability to display the entire memory acquisition on the screen.

3.23 X-Y Operation. The oscilloscope shall provide X-Y operation.

3.23.1 X-Axis and Impedance Range. The oscilloscope shall have an X-axis deflection factor, variable range, and accuracy as well as an input impedance range defined for channels 1 and 2.

3.23.2 X and Y Phase Difference. The oscilloscope phase difference between X and Y shall be less than 1° from DC to 1 MHz and less than 2° from 1 MHz to 2 MHz with probes provided.

3.24 Waveform, Setup, Reference and Calibration Data Storage. The oscilloscope shall provide internal waveform storage to store at least 4 waveforms at maximum memory storage, at least 4 reference waveforms, at least 4 front panel setups, and calibration data.

3.25 Back Up Battery. The oscilloscope shall have a back up battery that provides back up power for at least two years. The equipment shall have the capability to store front panel settings and to retain this data for a minimum of two years. If the equipment uses a rechargeable Nickel-Cadmium (Ni-Cad) battery, there shall be means by which the operator and/or internal circuitry can deep discharge the battery to prevent "memory effect."

3.26 Memory Data Transfer. The oscilloscope shall provide the capability to save any waveform or setup data to any applicable reference memory location, to transfer setup information and/or waveforms into a reference memory location via the GPIB interface, and to transfer setup and waveform data from the floppy disk drive.

3.27 Mass Storage. The oscilloscope shall provide mass storage providing additional storage of reference waveforms, setups, data files and image files. The oscilloscope may have a floppy disk in a 3.5-inch size, 1.4 Megabytes storage, and in a DOS format.

3.28 Alternate Mass Storage Devices. The oscilloscope may use a PCMCIA I/II/III flash memory card or other storage device with 1.4 Megabytes or greater capacity and DOS format.

3.29 Mass Storage Operational Restraints. The oscilloscope shall have the ability store, archive, retrieve, and any other storage operation over a temperature range of +10°C to +40°C.

3.30 Pass/Fail and Limit Testing. The oscilloscope shall have the capability to define limits and patterns for pass/fail and limit testing.

3.30.1 Waveform Failure Indication. The oscilloscope shall have the ability to indicate waveforms out of limits by the following means by use of: 1) waveform parameter limits or with "masks" which describe a waveform's shape, 2) a failure invoking an operator preprogrammed action, or 3) a combination of both 1) and 2).

3.30.2 Forms of Pre-Programmed Failure Indications. The oscilloscope shall display failures through its automated waveform measurement capability by: printing a hardcopy report of the failure; saving the failure in memory; responding to a GPIB PC computer-generated SRQ (service request); activating an audible alarm; or any combination of the preceding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

3.31 VGA Compatible Output. The oscilloscope shall have a VGA compatible output allowing an external VGA monitor to be connected to it by a D-type VGA connector.

4. REGULATORY REQUIREMENTS. In accordance with Section 23.403 of the Federal Acquisition Regulation, the federal government's policy is to acquire items composed of the highest percentage of recoverable materials practicable consistent with maintaining a satisfactory level of competition and without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials

5. PRODUCT CONFORMANCE.

5.1. Product Conformance. The oscilloscope provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met.

7.2 Exceeded Metric Products. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.3 Sources of Documents.

7.3.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094, Phone 215/697-2179

7.3.2 IEEE standards. A copy of IEE standards can be obtained from IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331. Phone: 800-678-4333 (Toll-Free, USA & Canada), 732-981-0060 (Worldwide), Fax: 732-981-9667.

7.3.3 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-F092)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at www.dodssp.daps.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:07Z
Check the source to verify that this is the current version before use.